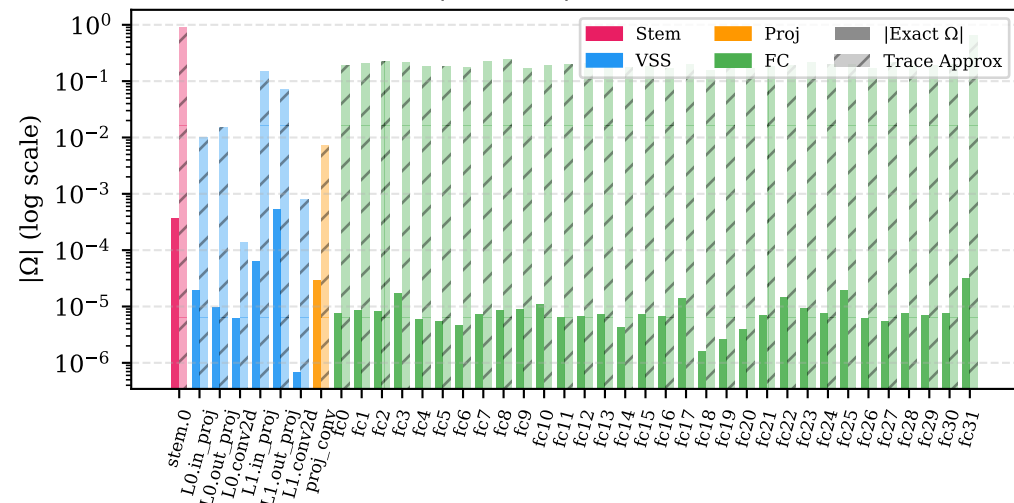
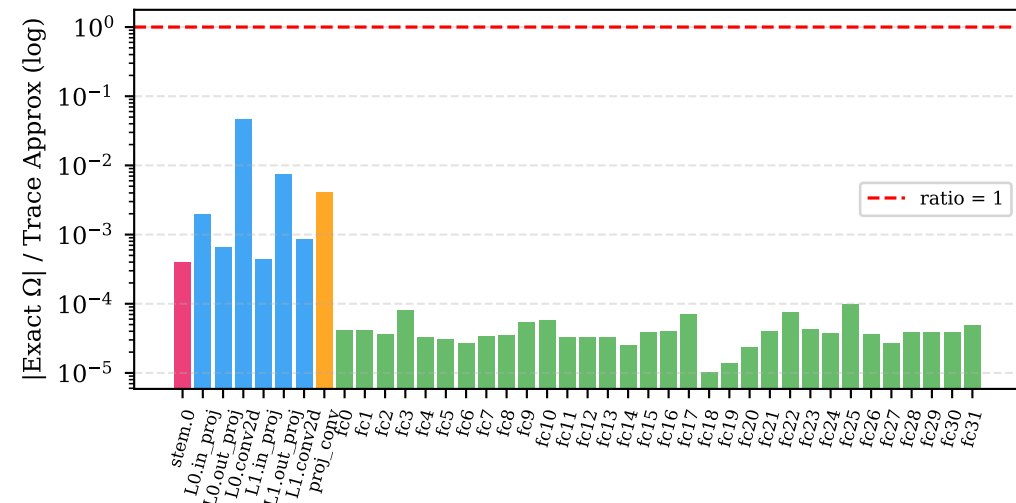


Lemma 1 Validation: Exact Block-Diagonal Ω vs Hutchinson Trace Approximation

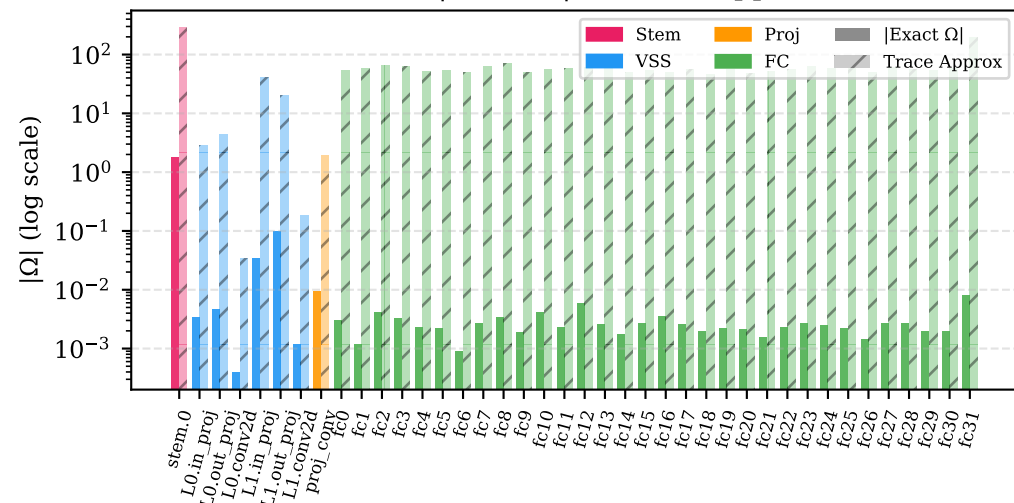
(INT8) |Exact Ω | vs Trace Approx



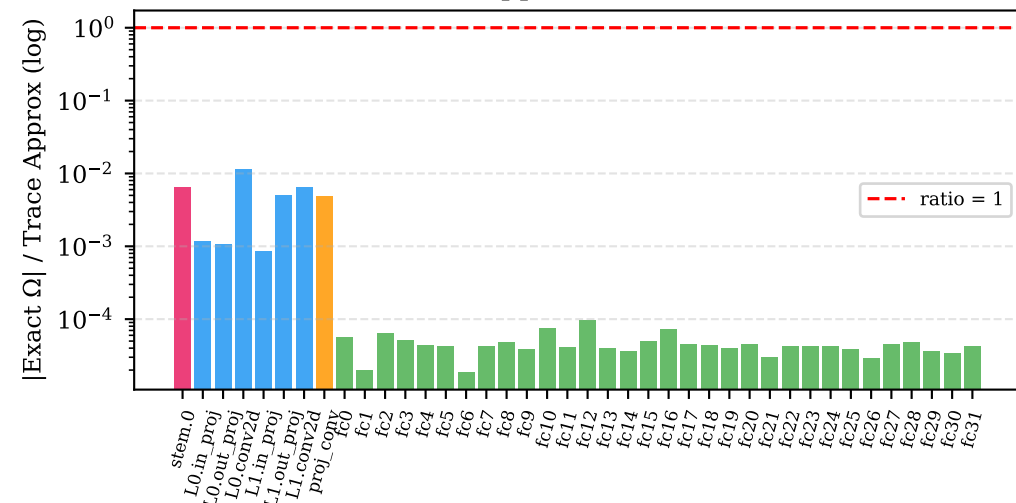
(INT8) Approximation Ratio



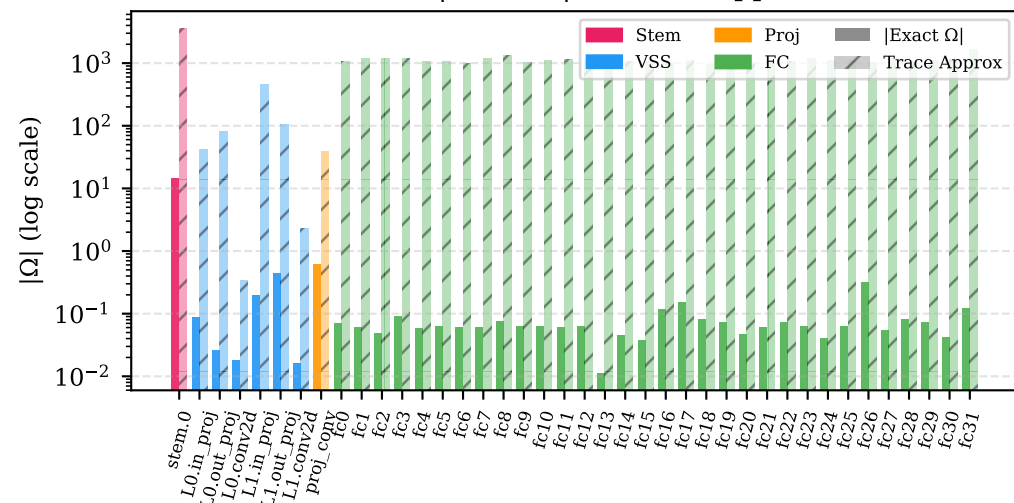
(INT4) |Exact Ω | vs Trace Approx



(INT4) Approximation Ratio



(INT2) |Exact Ω | vs Trace Approx



(INT2) Approximation Ratio

